

SCADS Software Requirements Specification

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1 Scope

1.1 Document Overview

This document specifies the software requirements for the Spacecraft Control Attitude Determination System (SCADS) v1.0. SCADS v1.0 is a portfolio demonstration project implementing a fault-tolerant finite state machine for attitude determination and control on a simulated rigid-body spaceship platform, running on bare-metal hardware under FreeRTOS.

1.2 Scope of v1.0

The v1.0 scope covers:

- Five-state operational finite state machine with formally defined transitions
- Fault detection for sensor dropout, out-of-range readings, task overruns, watchdog events, and estimator divergence
- Attitude estimation using a simplified estimator with characterized uncertainty growth
- Attitude control for detumble and safe modes; nominal-mode control is implemented as a stub that maintains zero commanded torque for attitude hold
- Ground command processing for a minimal command set
- Local event and state logging to non-volatile storage

1.3 Scope of v2.0 and Beyond

The following capabilities are deferred to v2.0 and beyond:

- Full attitude tracking control for arbitrary commanded attitudes
- Translational maneuver execution (thruster control)
- Ground telemetry packet generation and transmission
- Mission-time correlation and absolute time synchronization
- Multi-sensor fusion estimation (e.g., Extended Kalman Filter)
- Sensor redundancy and voting

Requirements related to deferred capabilities are not included in this document.

1.4 Related Documents

The Software Development Plan (SDP)

The System and Software Design Description (SSDD)

The Software Test Plan and Processing Procedures (STP)

2 Definitions and Acronyms

The table below contains definitions and acronyms used within the document.

Table 2.1: Definitions and Acronyms

Term	Definition
SCADS	Spacecraft Control Attitude Determination System
SDP	Software Development Plan
SSDD	System and Software Design Description
STPP	Software Test Plan and Procedures
TBD	To be determined
GPS	Global Positioning System
FreeRTOS	Free Real-Time Operating System

3 Functional Requirements

3.1 Required States

There exist required states of the spaceship. The finite state machine graph is visually displayed in the figure below, and the requirements are formalized in the subsequent section.

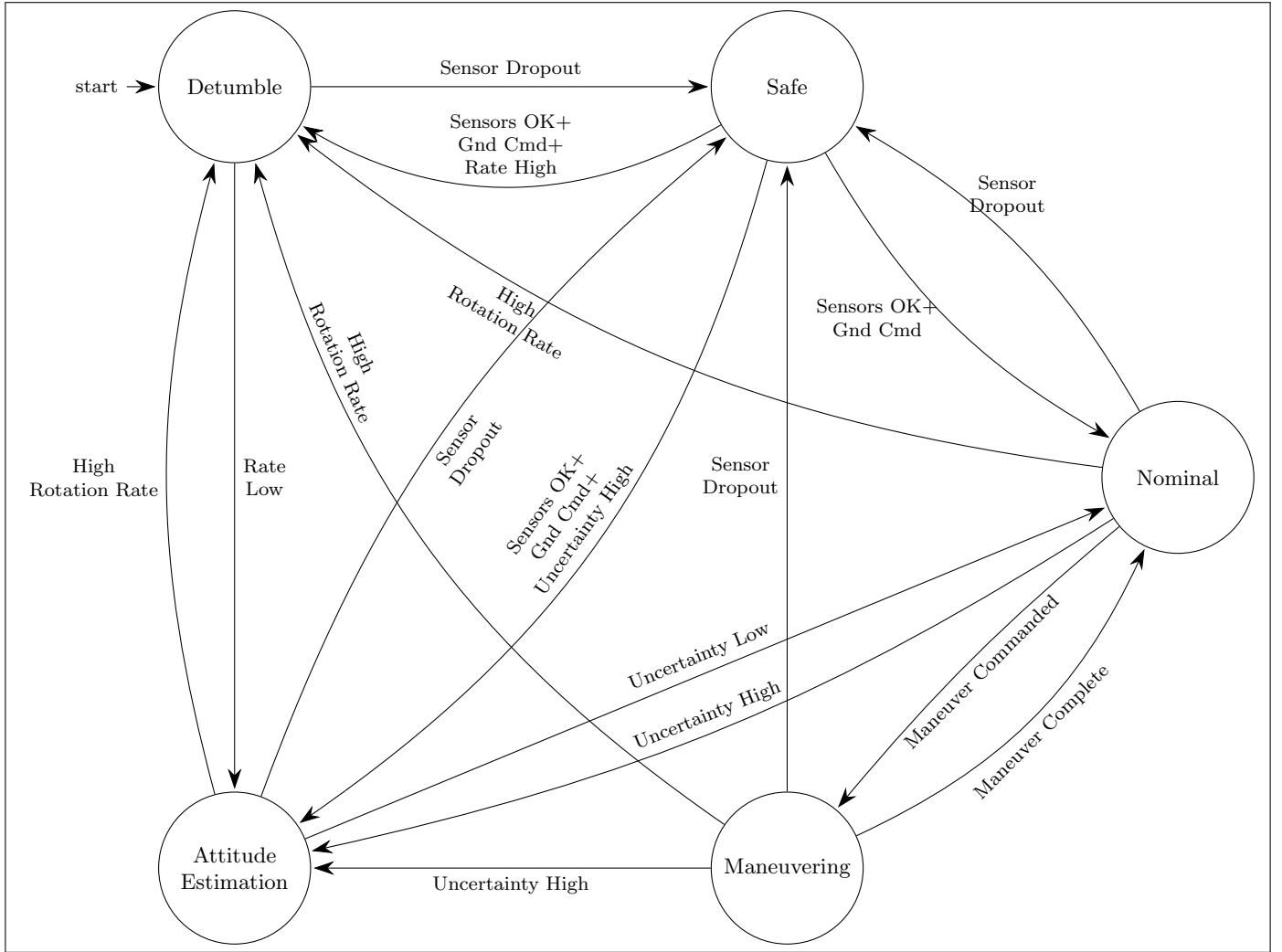


Figure 1: SCADS Finite State Machine

3.2 State Transition Requirements

The following table provides some common definitions used within the State Transition Requirements.

Table 3.1: Common Definitions

Phrase	Definition
Sensor Fault	Refers to a sensor going offline for SD_CONSECUTIVE_NUM_SAMPLES consecutive samples or SD_TRIGGER_AGGREGATE_SAMPLES samples per the last SD_AGGREGATE_SAMPLES.
Sensor Recover	Refers to a sensor coming back online for SR_CONSECUTIVE_NUM_SAMPLES consecutive samples or SR_TRIGGER_AGGREGATE_SAMPLES samples per the last SR_AGGREGATE_SAMPLES.
High Rotation	When the measured rate of rotation exceeds ROTATION_RATE_HIGH rad/s ² over ROTATION_SAMPLES_HIGH consecutive samples.
Low Rotation	When the measured rate of rotation is less than or equal to ROTATION_RATE_LOW rad/s over ROTATION_SAMPLES_LOW consecutive samples.
High Uncertainty	When the measured attitude uncertainty exceeds UNCERATINTY_HIGH units per UNCERTAINTY_SAMPLES_HIGH consecutive samples.
Low Uncertainty	When the measured attitude uncertainty is less than UNCERATINTY_LOW units per UNCERTAINTY_SAMPLES_LOW consecutive samples.

The following table defines the state transition requirements between the aforementioned states.

Table 3.2: State Transition Requirements

Requirement ID	Requirement	Acceptance Criteria
SCADS-ST-REQ-001	The SCADS shall transition from attitude estimation to nominal when the sensors indicate uncertainty low.	TBD - STPP
SCADS-ST-REQ-002	The SCADS shall transition from nominal to detumble with the highest precedence if the sensors indicate high rotation.	TBD - STPP
SCADS-ST-REQ-003	The SCADS shall transition from nominal to safe with the second highest precedence if a sensor faults.	TBD - STPP
SCADS-ST-REQ-004	The SCADS shall transition from nominal to attitude estimation with the third highest precedence if the sensors indicate high uncertainty.	TBD - STPP
SCADS-ST-REQ-005	The SCADS shall transition from nominal to maneuvering when a maneuver is commanded.	TBD - STPP
SCADS-ST-REQ-006	The SCADS shall transition from detumble to attitude estimation when the sensors indicate low rotation.	TBD - STPP
SCADS-ST-REQ-007	The SCADS shall transition from maneuvering to nominal when the commanded maneuver is completed.	TBD - STPP
SCADS-ST-REQ-008	The SCADS shall transition from attitude estimation to safe if a sensor faults.	TBD - STPP
SCADS-ST-REQ-009	The SCADS shall transition from maneuvering to detumble when sensors indicate high rotation.	TBD - STPP
SCADS-ST-REQ-010	The SCADS shall transition from detumble to safe when a sensor faults.	TBD - STPP
SCADS-ST-REQ-011	The SCADS shall transition from attitude estimation to detumble when sensors indicate high rotation.	TBD - STPP
SCADS-ST-REQ-012	The SCADS shall transition from maneuvering to attitude estimation when sensors indicate uncertainty high.	TBD - STPP

Requirement ID	Requirement	Acceptance Criteria
SCADS-ST-REQ-013	The SCADS shall transition from maneuvering to safe when a sensor faults.	TBD - STPP
SCADS-ST-REQ-014	The SCADS shall transition from safe to detumble with the highest precedence when sensors recover, a ground command is received, and sensors indicate high rotation.	TBD - STPP
SCADS-ST-REQ-015	The SCADS shall transition from safe to attitude estimation with the second highest precedence when sensors recover, a ground command is received, and sensors indicate uncertainty high.	TBD - STPP
SCADS-ST-REQ-016	The SCADS shall transition from safe to nominal with the third highest precedence when sensors recover, and a ground command is received.	TBD - STPP
SCADS-ST-REQ-017	The SCADS shall enter detumble when the system is powered on.	TBD - STPP
SCADS-ST-REQ-018	Upon any transition out of maneuvering prior to maneuver completion, SCADS shall command all thrusters to the off state.	TBD - STPP

3.3 Fault Detection Requirements

The fault detection requirements are defined in the table below.

Table 3.3: Fault Detection Requirements

Requirement ID	Requirement	Acceptance Criteria
SCADS-FD-REQ-001	The SCADS shall detect a Sensor Fault for each sensor, where Sensor Fault is defined as a sensor going off-line for SD_CONSECUTIVE_NUM_SAMPLES consecutive samples or SD_TRIGGER_AGGREGATE_SAMPLES samples per the last SD_AGGREGATE_SAMPLES.	TBD - STPP
SCADS-FD-REQ-002	The SCADS shall detect a sensor out-of-range fault for each sensor when any reading exceeds its specified minimum or maximum valid range [Sensor Specification, TBD].	TBD - STPP
SCADS-FD-REQ-003	The SCADS shall detect a task overrun fault when any critical task fails to complete execution within its allocated period.	TBD - STPP
SCADS-FD-REQ-004	The SCADS shall recognize a watchdog-induced reset condition upon boot and log it as a watchdog timeout fault.	TBD - STPP
SCADS-FD-REQ-005	The SCADS shall detect an estimator divergence fault when the attitude uncertainty exceeds UNCERTAINTY_DIVERGENCE_THRESHOLD or increases monotonically for DIVERGENCE_SAMPLES consecutive samples.	TBD - STPP
SCADS-FD-REQ-006	Upon detection of any fault defined in SCADS-FD-REQ-001 through SCADS-FD-REQ-005, the SCADS shall log the fault type, timestamp, and originating subsystem to non-volatile storage.	TBD - STPP

3.4 Core Functional Requirements

The following table outlines the core functional requirements of SCADS.

Table 3.4: Functional Requirements

Requirement ID	Requirement	Acceptance Criteria
SCADS-FR-REQ-001	The SCADS shall compute an attitude estimate at a rate of no less than ATTITUDE_ESTIMATE_RATE_HZ.	TBD - STPP
SCADS-FR-REQ-002	The attitude estimate shall consist of a quaternion representing spacecraft body-to-inertial orientation and an associated uncertainty value.	TBD - STPP
SCADS-FR-REQ-003	The attitude uncertainty shall be expressed in units of radians.	TBD - STPP
SCADS-FR-REQ-004	The attitude estimator shall make its estimates available for use to other spacecraft systems.	TBD - STPP
SCADS-FR-REQ-005	The SCADS shall compute actuator commands at a rate of no less than CONTROL_LOOP_RATE_HZ.	TBD - STPP
SCADS-FR-REQ-006	In detumble, the SCADS shall compute actuator commands that reduce spacecraft rotation rate.	TBD - STPP
SCADS-FR-REQ-007	In safe, the SCADS shall compute actuator commands that orient solar arrays toward the sun.	TBD - STPP
SCADS-FR-REQ-008	In nominal, the SCADS shall compute actuator commands that maintain the commanded attitude.	TBD - STPP
SCADS-FR-REQ-009	The SCADS shall acquire sensor readings at a rate of no less than SENSOR_ACQUISITION_RATE_HZ.	TBD - STPP
SCADS-FR-REQ-010	The SCADS shall convert raw sensor readings into physical units prior to use by other subsystems.	TBD - STPP
SCADS-FR-REQ-011	Each sensor reading shall be associated with a timestamp indicating the time of acquisition.	TBD - STPP
SCADS-FR-REQ-012	The SCADS shall issue actuator commands every control cycle, including zero-torque commands when no rotation is required.	TBD - STPP
SCADS-FR-REQ-013	Commanded actuator torques shall not exceed MAX_ACTUATOR_TORQUE.	TBD - STPP
SCADS-FR-REQ-014	The SCADS shall accept ground commands via the command interface defined in Section 5.	TBD - STPP
SCADS-FR-REQ-015	The SCADS shall validate each received command for format correctness and parameter range prior to execution.	TBD - STPP
SCADS-FR-REQ-016	The SCADS shall reject invalid commands and generate an error response.	TBD - STPP
SCADS-FR-REQ-017	The SCADS shall support the commands defined in the command dictionary [TBD: Appendix].	TBD - STPP
SCADS-FR-REQ-018	The SCADS shall track the operational status of each sensor.	TBD - STPP
SCADS-FR-REQ-019	The SCADS shall track the execution status of each critical task, including last execution time and missed deadline count.	TBD - STPP
SCADS-FR-REQ-020	Health status shall be made available to fault detection logic defined in Section 3.3.	TBD - STPP
SCADS-FR-REQ-021	The SCADS shall maintain a monotonic system time, initialized at boot, with a resolution of no less than SYSTEM_TIME_RESOLUTION.	TBD - STPP

Requirement ID	Requirement	Acceptance Criteria
SCADS-FR-REQ-022	The SCADS shall log the current operational mode, attitude estimate, and detected fault events to non-volatile storage.	TBD - STPP

4 Performance Requirements

The following section outlines the performance requirements.

Table 4.1: Performance Requirements

Requirement ID	Requirement	Acceptance Criteria
SCADS-P-REQ-001	From power on to first completion of one full control cycle, the SCADS shall complete the transition within <code>BOOT_TIME_SEC</code> seconds.	TBD - STPP
SCADS-P-REQ-002	From detection of a fault per Section 3.3, to entry to safe, the SCADS shall complete the transition in less than <code>SAFE_MODE_TRANS_MSEC</code> milliseconds.	TBD - STPP
SCADS-P-REQ-003	When in nominal the SCADS shall maintain attitude within <code>NOMINAL_ATTITUDE_VARIATION</code> radians of the attitude held at nominal entry.	TBD - STPP
SCADS-P-REQ-004	When in detumble the SCADS shall reduce overall rotation by no less than <code>DETUMBLE_DAMPENING_RATE</code> rad/s ² .	TBD - STPP
SCADS-P-REQ-005	The SCADS shall execute within <code>MAX_CPU_USAGE_PERCENT</code> percent CPU utilization on target hardware.	TBD - STPP
SCADS-P-REQ-006	The SCADS shall execute within <code>MAX_RAM_USAGE_PERCENT</code> percent RAM usage on the target hardware.	TBD - STPP
SCADS-P-REQ-007	From ground command receipt to dispatch, the SCADS shall complete processing within <code>GROUND_COMMAND_LATENCY_MAX</code> milliseconds.	TBD - STPP
SCADS-P-REQ-008	The SCADS shall acknowledge each ground command within <code>GROUND_ACKNOWLEDGEMENT_MAX</code> milliseconds.	TBD - STPP
SCADS-P-REQ-009	Each sensor acquisition cycle shall complete within <code>SENSOR_CYCLE_MAX_MS</code> milliseconds.	TBD - STPP
SCADS-P-REQ-010	When in attitude estimation the SCADS shall reduce attitude uncertainty from <code>UNCERTAINTY_HIGH</code> or greater to <code>UNCERTAINTY_LOW</code> within <code>ESTIMATOR_CONVERGENCE_MAX</code> seconds.	TBD - STPP
SCADS-P-REQ-011	Given an initial rotation rate not exceeding <code>DETUMBLE_MAX_INITIAL_RATE</code> rad/s the SCADS shall reduce the rotation rate to below <code>ROTATION_RATE_LOW</code> within <code>DETUMBLE_MAX_DURATION</code> seconds.	TBD - STPP
SCADS-P-REQ-012	Each control cycle, from start of computation to issuance of actuator commands, shall complete within <code>CONTROL_CYCLE_MAX_MS</code> milliseconds.	TBD - STPP

5 Interface Requirements

The table below outlines the interface requirements.

Table 5.1: Interface Requirements

Requirement ID	Requirement	Acceptance Criteria
SCADS-INT-REQ-001	The SCADS shall provide a ground command interface for receiving operational commands from a ground station or test fixture.	TBD - STPP
SCADS-INT-REQ-002	Ground commands shall conform to the message format specified in Appendix 8.	TBD - STPP
SCADS-INT-REQ-003	The SCADS shall provide a telemetry output interface for transmitting operational status.	TBD - STPP
SCADS-INT-REQ-004	The SCADS shall provide a sensor data acquisition interface returning three-axis angular rate, three-axis acceleration, and three-axis sun-vector readings.	TBD - STPP
SCADS-INT-REQ-005	The sensor data acquisition interface shall return readings with associated timestamps.	TBD - STPP
SCADS-INT-REQ-006	The SCADS shall provide an actuator command interface accepting three-axis torque commands.	TBD - STPP
SCADS-INT-REQ-007	All external interfaces shall handle malformed data by logging the event and discarding the affected message.	TBD - STPP

6 Safety and Fault Tolerance Requirements

The table below outlines the safety and fault tolerance requirements.

Table 6.1: Safety and Fault Tolerance Requirements

Requirement ID	Requirement	Acceptance Criteria
SCADS-SFT-REQ-001	Upon detection of any fault per Section 3.3, the SCADS shall transition to safe.	TBD - STPP
SCADS-SFT-REQ-002	In safe Mode, the SCADS shall command actuators to orient solar arrays toward the sun.	TBD - STPP
SCADS-SFT-REQ-003	In safe mode, the SCADS shall suspend all non-essential processing.	TBD - STPP
SCADS-SFT-REQ-004	The SCADS shall not autonomously exit safe mode under any condition.	TBD - STPP
SCADS-SFT-REQ-005	The SCADS shall log fault events and operational state changes for ground analysis upon entering safe mode.	TBD - STPP

7 Operational Environment Requirements

The table below outlines the operational environment requirements. Note v1.0 SCADS is validated for low earth orbit operation. The architecture is designed to support deep-space mission profiles in v2.0+ subject to environmental parameter updates.

Table 7.1: Operational Environment Requirements

Requirement ID	Requirement	Acceptance Criteria
SCADS-OE-REQ-001	The SCADS v1.0 is designed for operation in space environments with regular eclipse periods, microgravity, and thermal cycling consistent with orbital flight.	TBD - STPP
SCADS-OE-REQ-002	The SCADS shall operate over an internal electronics temperature range of MIN_OPERATING_TEMP°C to MAX_OPERATING_TEMP°C.	TBD - STPP
SCADS-OE-REQ-003	The SCADS architecture shall support adaptation to deep-space mission profiles (extended communication latency, no GPS availability, extended eclipse durations) in v2.0+.	TBD - STPP

8 Appendix

Table 8.1: Command Definitions

Command ID	Description	Format
SAFE_TO_NOMINAL	The command sent to transition from safe to nominal.	TBD - STPP
SAFE_TO_DETUMBLE	The command sent to transition from safe to detumble.	TBD - STPP
SAFE_TO_ESTIMATION	The command sent to transition from safe to attitude estimation.	TBD - STPP